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PROJECT NOTES: AUTOMATIC TRAFFIC LIGHT

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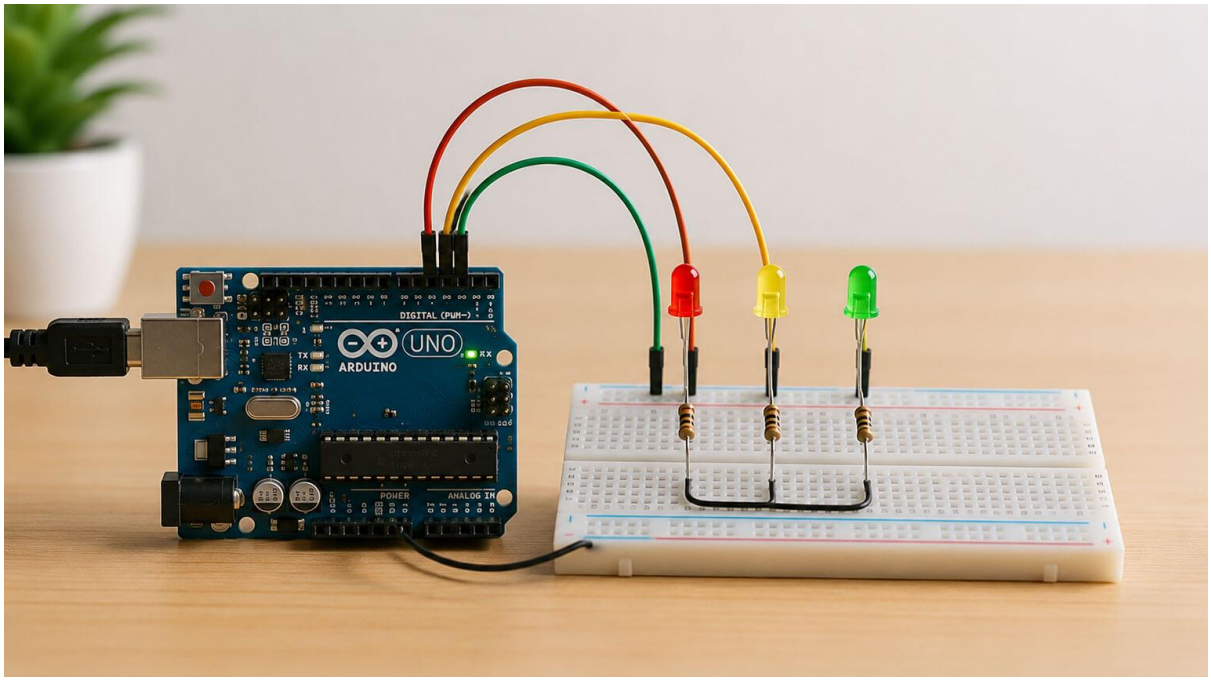
1. PROJECT OVERVIEW

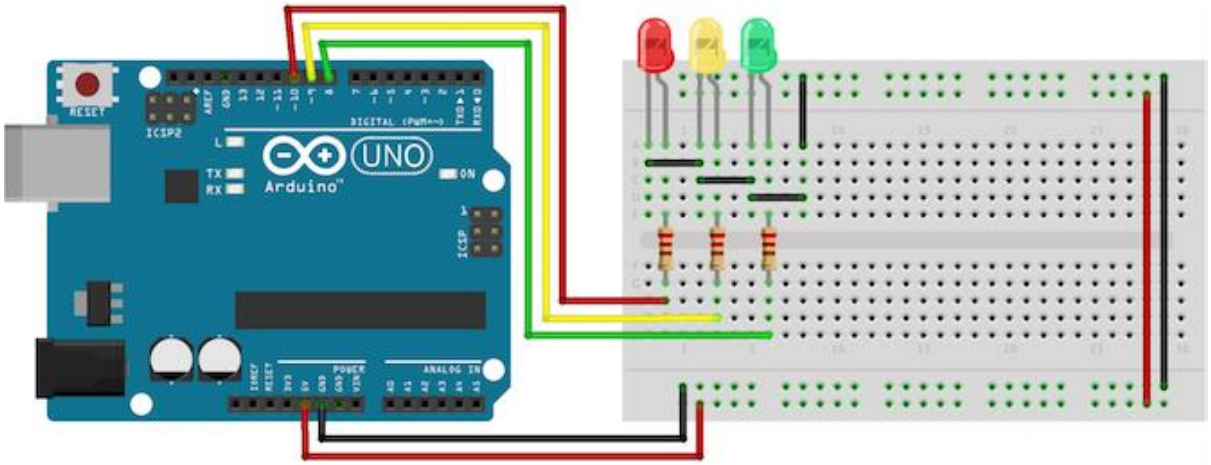
This project creates an automatic traffic light system. It uses an Arduino Uno to control three LEDs (Red, Yellow, Green) in a timed loop. This mimics a real street intersection signal.

2. COMPONENTS REQUIRED & THEIR USES

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* Arduino Uno: The microcontroller board. It acts as the "brain" of the circuit, executing code to control electricity flow.
* 3 LEDs (Red, Yellow, Green): Light Emitting Diodes. They represent the traffic signals (Stop, Ready, Go).
* 3 Resistors (220 ohms): Current-limiting components. They protect the LEDs from drawing too much current and burning out.
* Breadboard: A solderless grid board. It allows easy connection of components and jumper wires.
* Jumper Wires: Conducting wires used to bridge electrical connections between the Arduino and the breadboard.





3. HARDWARE STEP-BY-STEP GUIDELINES

Follow these steps to assemble the circuit on the breadboard:

[Step 1] Place the LEDs

* Insert the Red, Yellow, and Green LEDs onto separate rows of the breadboard.

* Note: Each LED has a longer leg (Anode/Positive +) and a shorter leg (Cathode/Negative -).

[Step 2] Connect the Ground Rail

* Run a jumper wire from the "GND" pin on the Arduino to the negative (-) rail (the long side strip) of the breadboard.

[Step 3] Add Resistors

* For each LED, connect a 220-ohm resistor from its shorter leg (Cathode) to the negative (-) rail of the breadboard. This provides a clear ground path for all 3 lights.

[Step 4] Connect Digital Pins

* Connect a jumper wire from the longer leg (Anode) of the Red LED to Digital Pin 10 on the Arduino.
* Connect a jumper wire from the longer leg (Anode) of the Yellow LED to Digital Pin 9 on the Arduino.
* Connect a jumper wire from the longer leg (Anode) of the Green LED to Digital Pin 8 on the Arduino.

4. ARDUINO PROGRAM CODE

Type this exact code into your Arduino IDE and upload it to your board:

```
// Define the digital pins for each LED
int redLed = 10;
int yellowLed = 9;
int greenLed = 8;

void setup() {
  // Set all LED pins as outputs
  pinMode(redLed, OUTPUT);
  pinMode(yellowLed, OUTPUT);
  pinMode(greenLed, OUTPUT);
```

```
}

void loop() {
  // 1. Red Light ON (Green and Yellow OFF) - Cars Stop
  digitalWrite(redLed, HIGH);
  digitalWrite(yellowLed, LOW);
  digitalWrite(greenLed, LOW);
  delay(5000); // Wait 5 seconds

  // 2. Green Light ON (Red and Yellow OFF) - Cars Go
  digitalWrite(redLed, LOW);
  digitalWrite(yellowLed, LOW);
  digitalWrite(greenLed, HIGH);
  delay(5000); // Wait 5 seconds

  // 3. Yellow Light ON (Red and Green OFF) - Cars Slow Down
  digitalWrite(redLed, LOW);
  digitalWrite(yellowLed, HIGH);
  digitalWrite(greenLed, LOW);
  delay(2000); // Wait 2 seconds
}
```

5. OPERATIONAL SEQUENCE SUMMARY

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* Stage 1: Red LED lights up for 5000 milliseconds (5 seconds).
* Stage 2: Red turns off, Green LED lights up for 5000 milliseconds (5 seconds).
* Stage 3: Green turns off, Yellow LED lights up for 2000 milliseconds (2 seconds).
* The cycle automatically repeats indefinitely.